

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5113

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks: 100

Annexure A

CONCEPT MAPPING

Code of Conduct:

I DINDU PUJITHA(192373037) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D PUJITHA

Annexure A

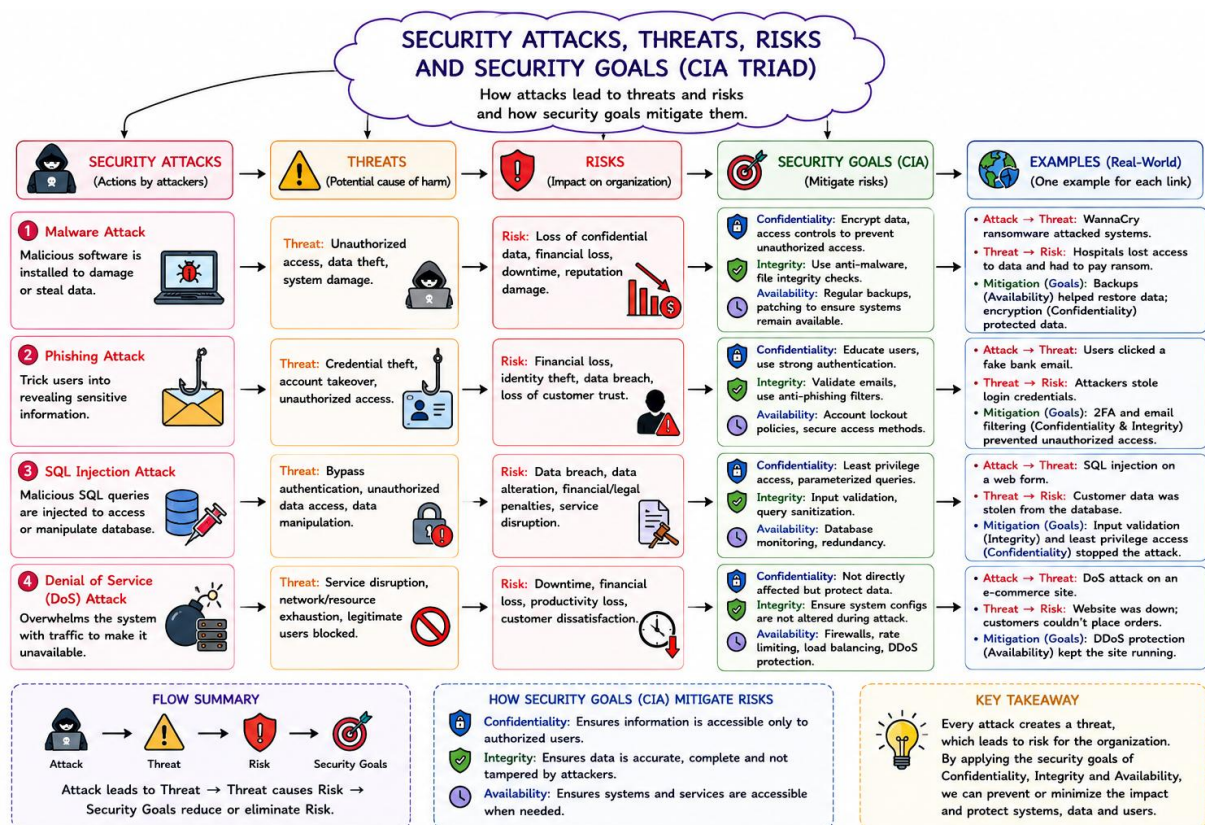
CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:

- Show how each attack leads to specific threats and risks
- Map how security goals help mitigate them Provide one real-world example for each link.

Answer:

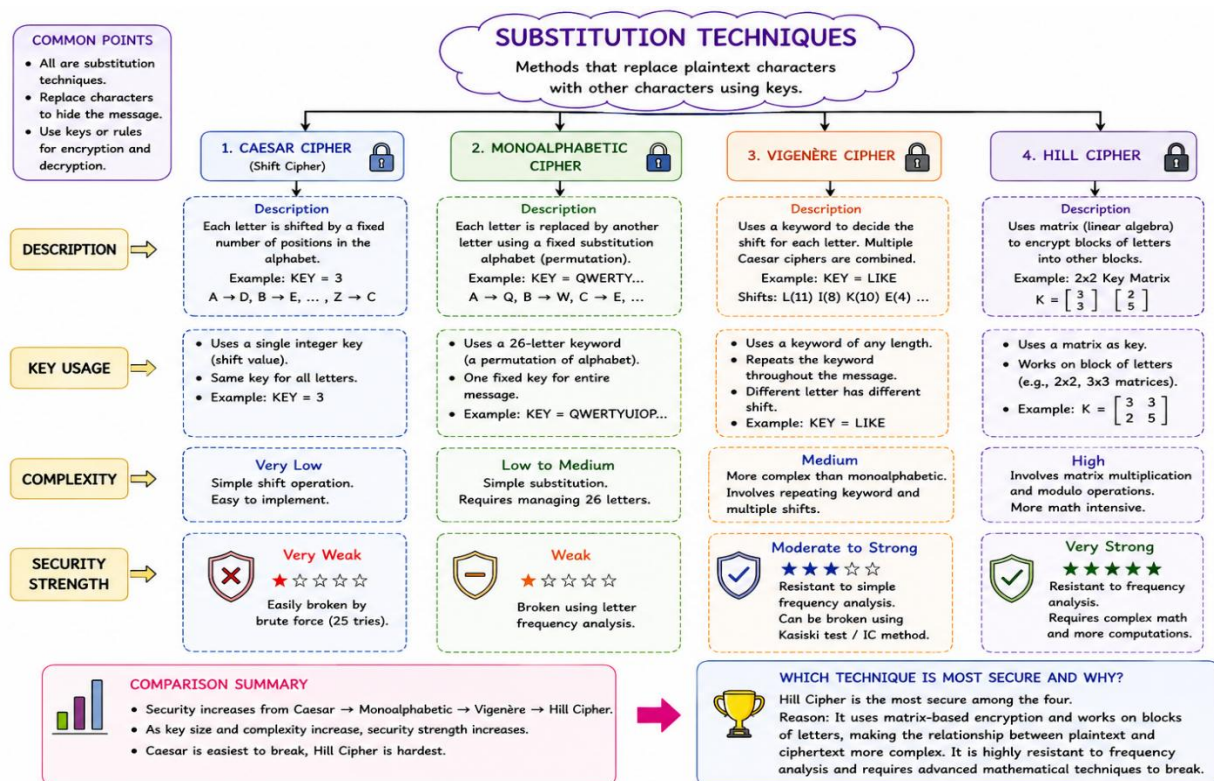
Concept Map:



2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

Answer:

Concept map:



3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

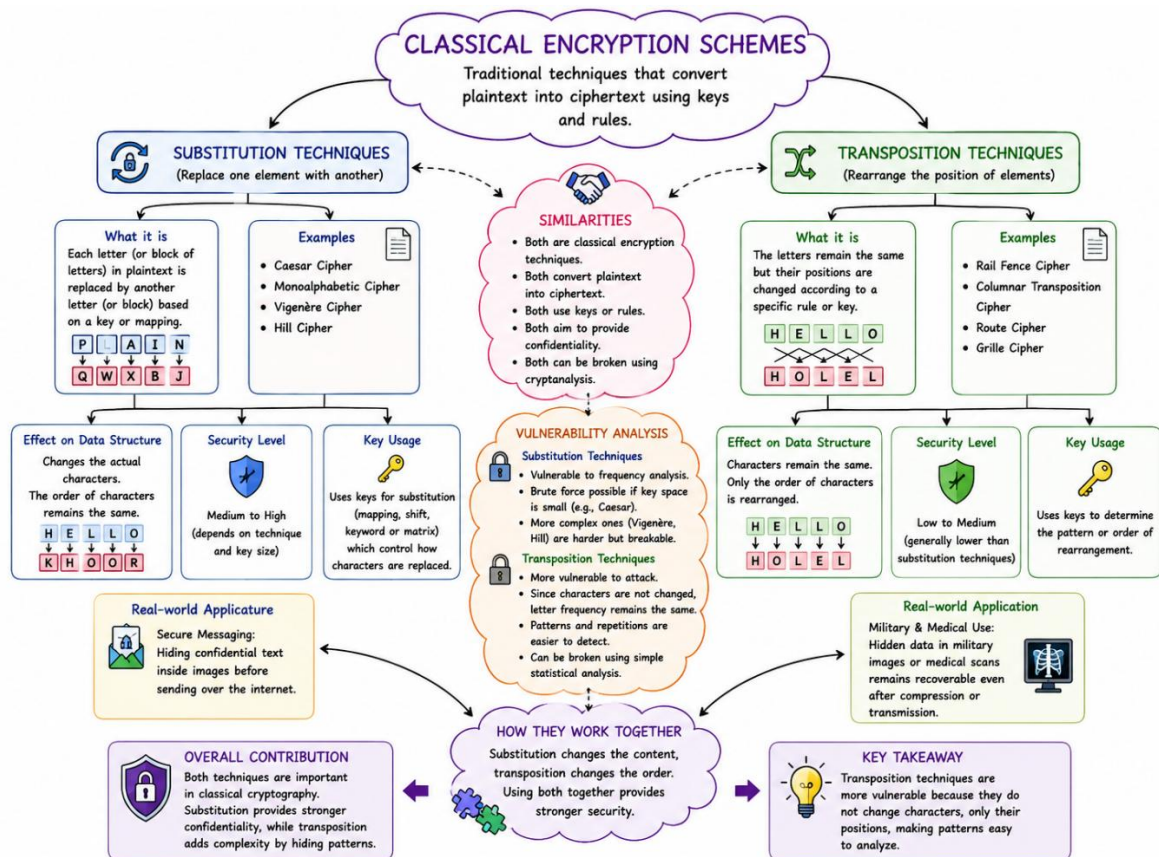
a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyze which is more vulnerable to attacks and why

Answer:

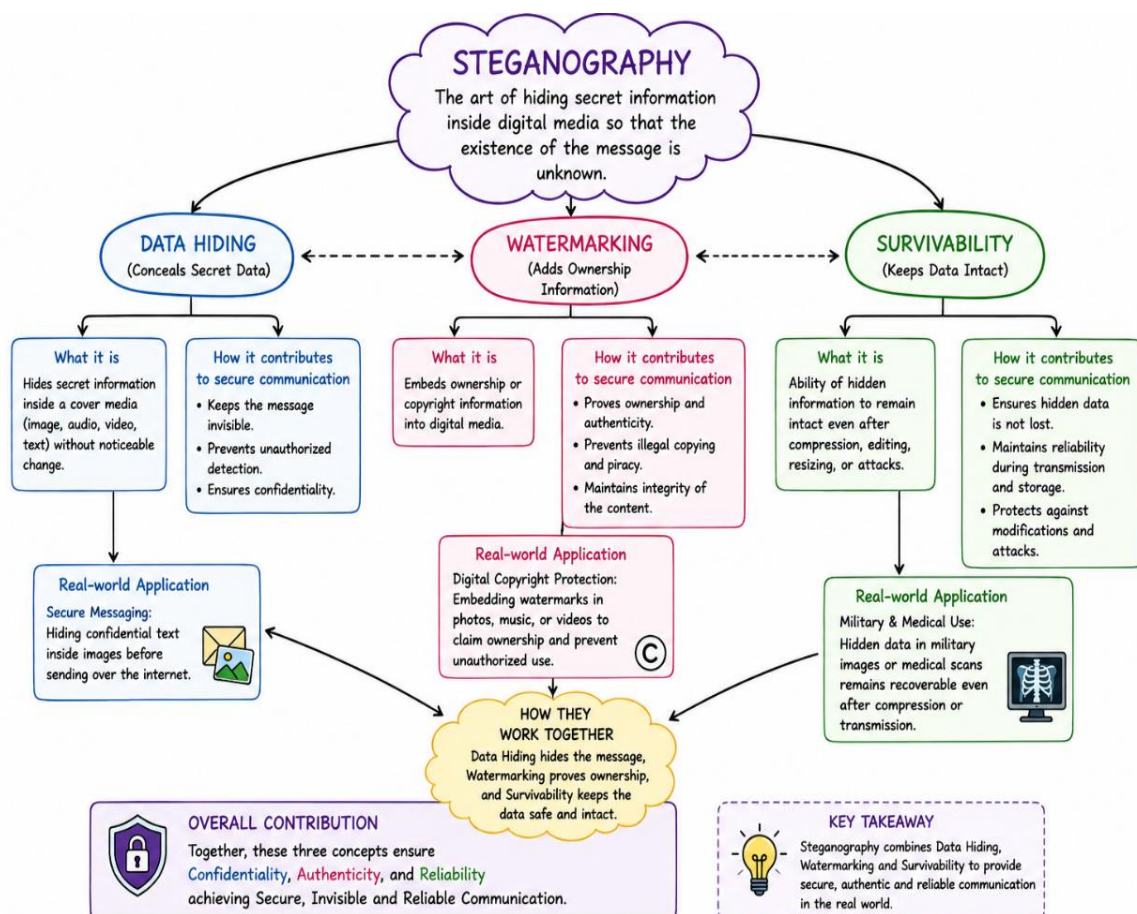
Concept map:



4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
- Show relationships between these concepts
 - Analyze how each contributes to secure communication
 - Include one real-world application for each

Answer:

Concept Map:



5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

- Show how each concept is mathematically connected
- Map their role in encryption/decryption
- Analyze which concept is most critical for public key cryptography and justify

Answer:

Concept Map:

